

2022 Underground Pipeline Bigdata Challenge

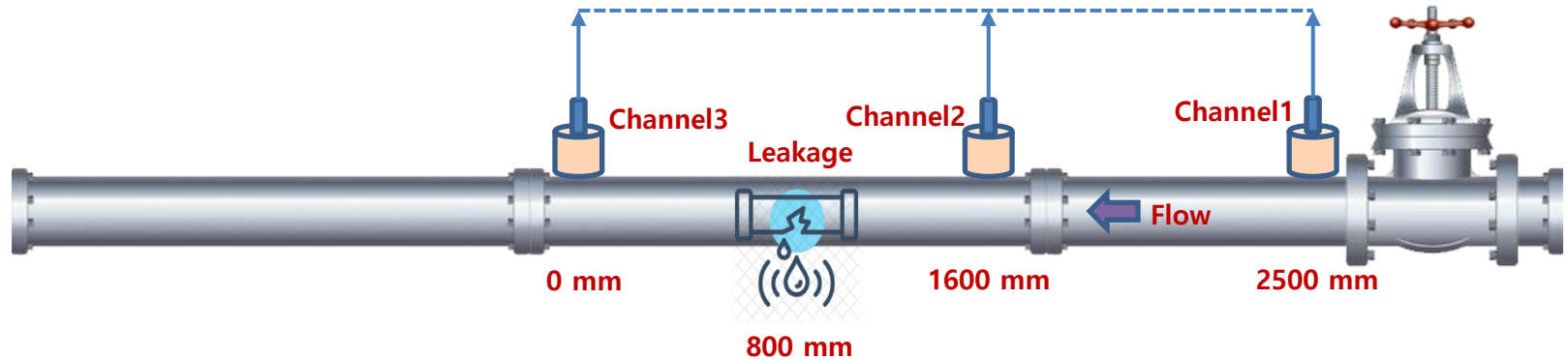
Preliminary round : 2022.10.31.~2021.11.21.

Final round : 2022.11.25.



- **Caution : All dataset has been copyrighted. Thus, use only for the challenge and do not share it to anyone and anywhere.**

Goals



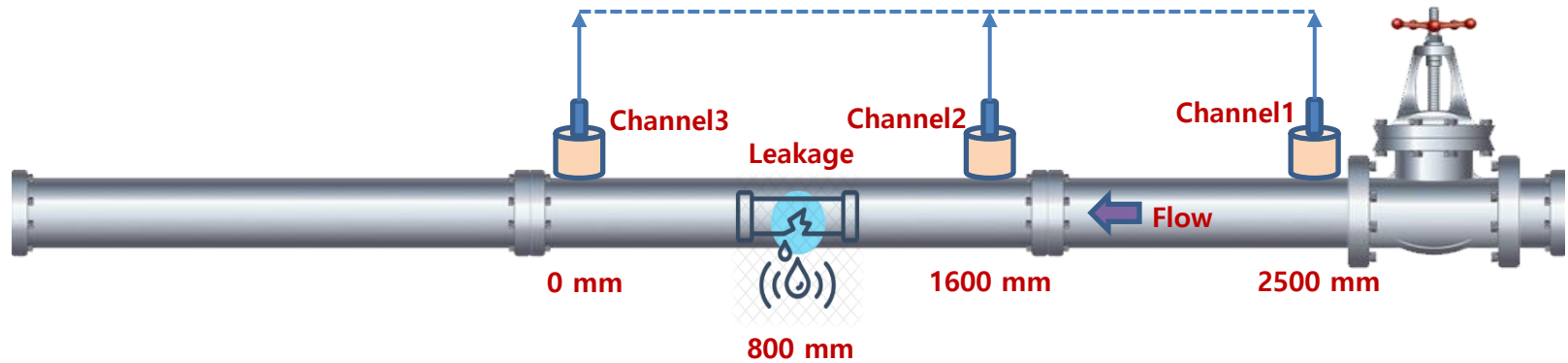
Leakage Detection

- ❖ Classify leaked and non-leaked condition in dataset

Leakage Localization

- ❖ Estimate the location of leakage in pipeline

Training dataset



- Sensor Type: Acoustic Emission Sensor
- Sampling Rate: 1MHz
- Total length of signal: 360 (s)
- Position of channel1: 2500mm
- Position of channel2: 1600mm
- Position of channel3: 0mm
- Leak location: 800mm
- Pinhole size of leak: 1mm
- Pressure: 13Bar, 18Bar
- Substance : Gas, Water
- Material of Pipeline : Steel

Download : <http://203.250.78.96:8000/sharing/R6TY4dvni>

Password : ITFault2022

Period : 2022.10.31 ~ 2022.11.21

Evaluation

Leak detection accuracy and location estimation error rate will be evaluated using three evaluation dataset

(However, the same algorithm and learning model are used in all steps)

First dataset

- Sensor Type: Acoustic Emission Sensor
- Sampling Rate: 1MHz
- Total length of signal: 360 (s)
- Position of channel1: 2500mm
- Position of channel2: 1600mm
- Position of channel3: 0mm
- Leak location: **revealed later**
- Pinhole size of leak: 1mm
- Pressure: 7Bar
- Substance : Water
- Material of Pipeline : Steel

Note : **Shared in 2022.11.15**

Download : <http://203.250.78.96:8000/sharing/Algplgk0>

Password : ITFault2022

Evaluation

Leak detection accuracy and location estimation error rate will be evaluated using three evaluation dataset

(However, the same algorithm and learning model are used in all steps)

Second dataset

- Sensor Type: Acoustic Emission Sensor
- Sampling Rate: 1MHz
- Total length of signal: 360 (s)
- Position of channel1: 2500mm
- Position of channel2: 1600mm
- Position of channel3: 0mm
- Leak location: **revealed later**
- Pinhole size of leak: 0.5mm
- Pressure: 7Bar
- Substance : Gas
- Material of Pipeline : Steel

Note : **Shared in 2022.11.15**

Download : <http://203.250.78.96:8000/sharing/Algplgk0>

Password : ITFault2022

Evaluation

Leak detection accuracy and location estimation error rate will be evaluated using three evaluation dataset

(However, the same algorithm and learning model are used in all steps)

Third dataset

- Sensor Type: Acoustic Emission Sensor
- Sampling Rate: 1MHz
- Total length of signal: 95 (s)
- Position of channel1: 0mm
- Position of channel2: 60000mm
- Position of channel3: 120000mm
- Leak location: **revealed later**
- Pinhole size of leak: 0.7mm
- Pressure: 10Bar
- Substance : Water
- Material of Pipeline : Steel

Note : **Shared in 2022.11.15**

Download : <http://203.250.78.96:8000/sharing/Algplgk0>

Password : ITFault2022

Evaluation

Leak detection accuracy and location estimation error rate will be evaluated using three evaluation dataset

(However, the same algorithm and learning model are used in all steps)

Evaluation

- Accuracy (based on confusion matrix)

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

- Localization error

$$L_e = \frac{1}{N} \sum_{t=1}^N \frac{|L_p(t) - L_a|}{D_{max}}$$

L_e : Average error rate of Localization

$L_p(t)$: Estimated leak location in t seconds

L_a : Actual leak location

D_{max} : Maximum distance between sensors

N : Total signal length in second unit

- Submit result (see CSV attachment)
 - Leak or not for each second
 - Estimated location for each second

Submission

- Data analysis result file (by November 21st)
 - Leak or not for each second
 - Estimated location for each second
- Algorithm source code (by November 21st)
- Learned artificial intelligence model file (by November 21st)
- Report (by November 21st)
- Algorithm description presentation material (ppt file, flowchart, data analysis content, description and source of the algorithm used, etc., by November 23)
- More materials (initial source code, description, csv form) in
Download : <http://203.250.78.96:8000/sharing/R6TY4dvni>
Password : ITFault2022
Period : 2022.10.31 ~ 2022.11.21



If you have any questions, contact at kjy7097@gmail.com